

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Introduction to Differential Calculus

Week 6 Day 1 AM



OUTLINE

① Graphs and Derivatives

② Optimization and Derivatives

③ Summary

INCREASING AND DECREASING

Recall that geometrically, $f'(x_0)$ is the slope of the tangent line to $y = f(x)$ at x_0 .

- If $f'(x_0) > 0$, the slope at x_0 is positive, i.e. the instantaneous rate of change of f w.r.t x is positive at x_0 , then $f(x)$ is increasing at x_0 .
- If $f'(x_0) < 0$, the slope at x_0 is negative, i.e. the instantaneous rate of change of f w.r.t x is negative at x_0 , then $f(x)$ is decreasing at x_0 .
- $f'(x_0) = 0$, the slope at x_0 is 0 and the instantaneous rate of change of f w.r.t x is 0 at x_0 . However, it's inconclusive about the growth of f at x_0 , it can be increasing, decreasing or constant.

For instance, $f(x) = x^3$ has zero derivative at 0 but it's increasing there. On the other hand, $g(x) = -x^3$ has zero derivative at 0 but it's decreasing there. The function $h(x) = x^2$ has zero derivative at 0 but it's neither increasing nor decreasing at $x = 0$, in fact, it's a local and global minimum there.

GRAPHS AND DERIVATIVES

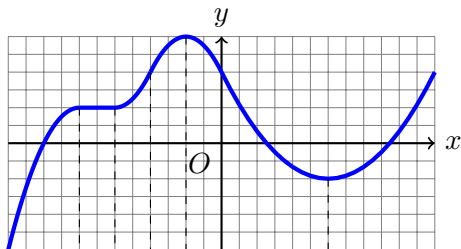
Thus, by knowing the sign of the derivative of a differentiable function, we can sketch its graph.

You can use this [Geogebra applet](#) to see how the slope (the value b) changes as you move along the graph of a function. Feel free to experiment it with different functions.

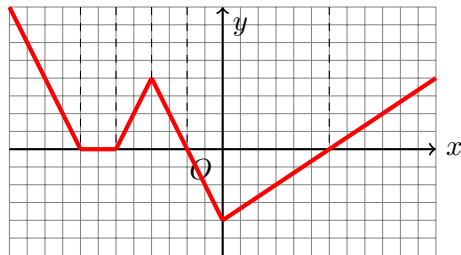
On the other hand, if you have the graph of a function f , you can derive qualitatively the graph of f' :

- $f'(x)$ is positive when the graph of f is increasing at x .
- $f'(x)$ is negative when the graph of f is decreasing at x .
- $f'(x)$ is zero when the graph of f is horizontal at x .
- The magnitude f' will get larger if the graph of f becomes steeper.

GRAPHING THE DERIVATIVE OF A FUNCTION



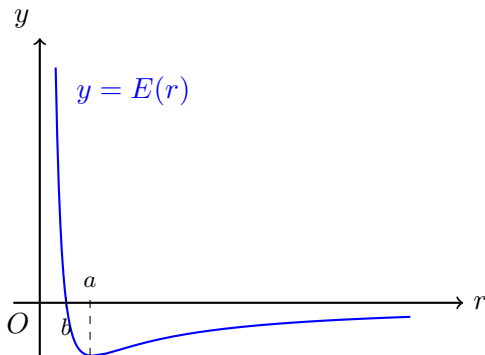
$$y = f(x)$$



$$y = f'(x)$$

ACTIVITY 1

The following graph shows how the potential energy $E(r)$ for a diatomic molecule changes as a function of the internuclear distance r . The definition of the force of interaction between two atoms is $F = -\frac{dE}{dr}$.

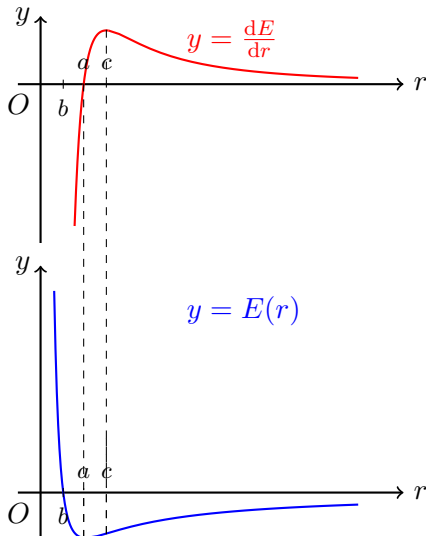


ACTIVITY 1

- (a) Sketch the graph of $\frac{dE}{dr}$ versus r . Indicate clearly how $\frac{dE}{dr}$ is changing with the curvature of the potential energy graph.
- (b) Which point on the graph of $\frac{dE}{dr}$ corresponds to the minimum of the potential energy curve? Explain your answer.
- (c) Now, sketch the graph of force of interaction F that explains the bond formation between two atoms.

ACTIVITY 1: SOLUTION

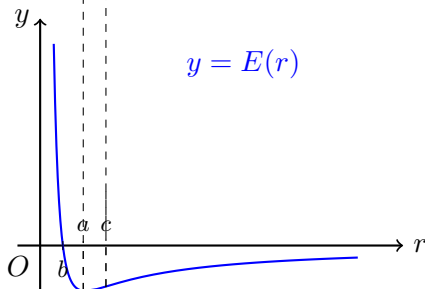
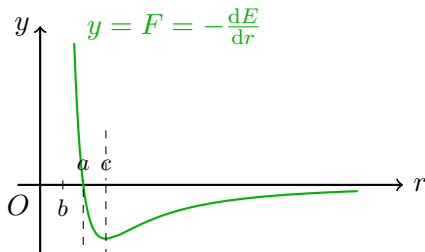
(a), (b)



- The slope of $y = E$ is getting less and less negative when r changes from 0 to a .
- The slope of $y = E$ is 0 at $r = a$.
- The slope of $y = E$ is positive and increasing for $a < r < c$.
- For $r > c$, the slope of E is positive, decreasing and approaches 0 as r approaches ∞ .
- The x -intercept $(a, 0)$ of the graph $y = \frac{dE}{dr}$ corresponds to the min of E .

ACTIVITY 1: SOLUTION

(c)



- The graph of $y = F = -\frac{dE}{dr}$ is obtained by reflecting the graph $y = \frac{dE}{dr}$ about the x -axis.

GEOMETRICAL RELATIONSHIPS OF f , f' AND f''

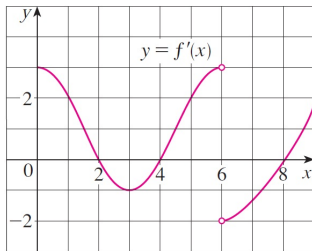
The following holds for **differentiable** functions $f(x)$ over an **interval** I .

- f is **non-decreasing over** I if and only if $f'(x)$ is non-negative, i.e. $f'(x) \geq 0$ for all $x \in I$.
- f is **non-increasing over** I if and only if $f'(x)$ is non-positive, i.e. $f'(x) \leq 0$ for all $x \in I$.
- If f has a local maximum or a local minimum at a , then $f'(a) = 0$.
- f is **convex over** I if and only if $f'(x)$ is **non-decreasing over** I .
- If f'' exists, f is **convex over** I if and only if $f''(x) \geq 0$ for all $x \in I$.
- f is **concave over** I if and only if $f'(x)$ is **non-increasing over** I .
- If f'' exists, f is **concave over** I if and only if $f''(x) \leq 0$ for all $x \in I$.

You don't have to memorize these results if you understand the meaning of f' and f'' .

ACTIVITY 2

The graph of **the derivative** of a continuous function f with domain $[0, 9]$ is shown below.



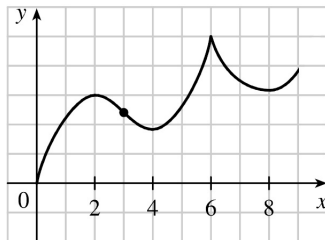
- On what intervals is f increasing? Decreasing?
- On what intervals is f convex? Concave?
- State the x -coordinate(s) of the inflection point(s).
- Given that $f(0) = 0$, sketch the graph of f .
- At what values of x does f have a local maximum? Local minimum?

ACTIVITY 2: SOLUTION

- (a) Since $f'(x) > 0$ over the intervals $(0, 2)$, $(4, 6)$ and $(8, 9)$, so f is increasing over these intervals. Similarly, $f'(x) < 0$ over the intervals $(2, 4)$ and $(6, 8)$, so f is decreasing over these intervals.
- (b) Since the slope of the tangent to the graph $f'(x)$ is $f''(x)$, we see that $f''(x) \leq 0$ over $[0, 3]$ and $f''(x) \geq 0$ over $[3, 6) \cup (6, 9]$ ($f''(6)$ is undefined because f' is discontinuous at $x = 6$). Thus, f is concave over $[0, 3]$ and convex over $[3, 6)$ and $(6, 9]$.
- (c) There's only one change in convexity at $x = 3$, so the only inflection point is at $x = 3$.

ACTIVITY 2: SOLUTION

- (d) We sketch the graph of f by using the information obtained from the previous parts:



- (e) From the graph, we can see that f has local maxima at $x = 2, 6$ and 9 ; local minima at $x = 0, 4$ and 8 .

OUTLINE

- ① Graphs and Derivatives
- ② Optimization and Derivatives
- ③ Summary

OPTIMIZATION PROBLEMS

An optimization problem is a problem of finding the global maximum and/or global minimum values of a function $f : D \rightarrow \mathbb{R}$.

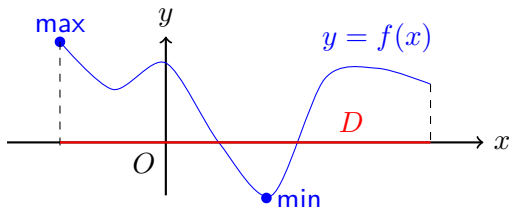
We often encounter optimization problems in real life: maximizing profits, distance traveled, areas, volumes; minimizing errors, costs, production/traveling time, space occupied.

In this section, we'll learn how to apply differentiation to solve optimization problems.

FORMULATION OF OPTIMIZATION PROBLEMS

The general strategy to solve an optimization problem is to first properly formulate it as a mathematical statement.

1. We express the value y that we want to optimize as a function (called the **objective function**) of the free variable/parameter x (called the **decision variable**) that we are allowed to vary: $y = f(x)$.
2. We also need to find out the set of values of x that are feasible/allowed (called the **feasible region** or the domain), say $x \in D$.
3. With this information, the optimization problem is simply to find the largest and/or smallest values of $y = f(x)$ for $x \in D$.



FERMAT'S THEOREM

FERMAT'S THEOREM

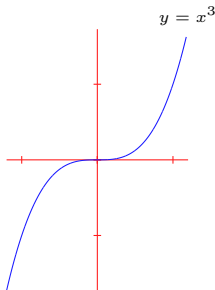
If $(x^*, f(x^*))$ is a local maximum/minimum and $f'(x^*)$ exists, then $f'(x^*) = 0$.

We'll omit the proof here.

Remarks:

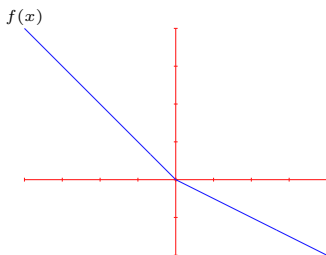
- The converse of the theorem is not true, i.e. $f'(x^*) = 0$ does **not** imply that $(x^*, f(x^*))$ is a local maximum or local minimum. For example, $f(x) = x^3$.

$f'(0) = 0$, but $(0, 0)$ is neither a local maximum nor minimum.

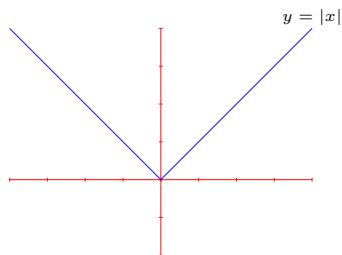


REMARKS ON FERMAT'S THEOREM

- Be careful if $f'(x^*)$ does not exist.



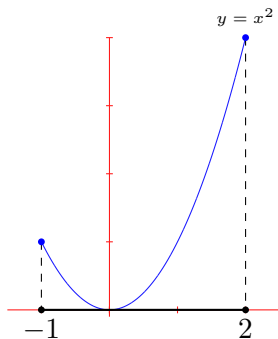
$f'(0)$ does not exist and $(0,0)$ is not a local maximum nor minimum.



$f'(0)$ does not exist and $(0,0)$ is a local and global minimum.

REMARKS ON FERMAT'S THEOREM

- Beware of endpoints.



Sometimes we want to find the extrema of a differentiable function $f : [a, b] \rightarrow \mathbb{R}$ over the closed interval $[a, b]$ with a, b as the endpoints. It's possible that the extrema occur at the endpoints even the one-sided derivative is not zero there.

For $f(x) = x^2$ over $[-1, 2]$, the global maximum is at $x = 2$, which is an endpoint and $f'(2) \neq 0$.

EXTREME VALUE THEOREM

EXTREME VALUE THEOREM (EVT)

If $f(x)$ is **continuous** on a **bounded closed interval** $[a, b]$, then f attains its maximum value and its minimum value at respective points in $[a, b]$.

Remarks:

- We say that an interval $I \subseteq \mathbb{R}$ is **bounded** if it's of finite length, i.e. there exists a real number $C > 0$ such that $I \subseteq [-C, C]$. More generally, a subset $D \subseteq \mathbb{R}$ is called **bounded** if there exists a real number $C > 0$ such that $D \subseteq [-C, C]$.
- Note that not all functions have a global maximum or a global minimum. The essence of the EVT is that it guarantees the existence of global maximum and global minimum for functions that satisfy the conditions: continuous with a bounded closed interval as its domain.
- If the domain of f is not a bounded closed interval, we can't apply the EVT and further analysis is required.

CRITICAL POINTS

How do we find the global maximum/minimum of $f(x)$ over $[a, b]$?
If f is continuous over $[a, b]$, the EVT guarantees the existence of global maximum/minimum and we just need to narrow down the search using Fermat's theorem.

PROPOSITION

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, then the candidates, x^* , at which the global maximum/minimum occur are

- $x^* \in [a, b]$ where $f'(x^*) = 0$.
- $x^* \in [a, b]$ where $f'(x^*)$ does not exist.
- $x^* = a$ and $x^* = b$.

They are called the **critical points** of f .

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous over $[a, b]$, all we need to do is just to find all the critical points x^* of f and then compute and compare the values of $f(x^*)$. The largest is the global maximum value and the smallest is the global minimum value.

SOLVING OPTIMIZATION PROBLEMS BY THE EVT - EXAMPLE

E.g.: Queen Dido's problem (814 BC).

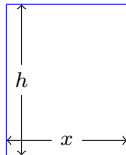
The queen fled to the African coast after her husband was killed. She begged King Iarbas, the local ruler for as much land as could be enclosed by a bull's hide. He agreed. She sliced the hide into small strips and used it to surround a large area. Her problem is known today as the *isoperimetric problem*:

Determine a plane figure of the largest possible area whose boundary has a specific fixed length.

QUEEN DIDO'S PROBLEM

We need some assumptions on the shape of the area, otherwise this is a very difficult optimization problem that requires the advanced topic called calculus of variations. Let's assume the shape of the area is a rectangle, with side lengths x and h , respectively. Let the total length of the hide be c , then $2x + 2h = c$.

We want to find the maximum possible area xh .



Problem: Maximize xh , where $x + h = \frac{c}{2}$ and $0 \leq x, h \leq \frac{c}{2}$.

Change this into a problem with just one decision variable by substituting $h = \frac{c}{2} - x$.

Problem: Maximize $f(x) := x(\frac{c}{2} - x)$, where $x \in [0, \frac{c}{2}]$.

SOLVING QUEEN DIDO'S PROBLEM

Problem: Maximize $f(x) = x \left(\frac{c}{2} - x \right)$, where $x \in \left[0, \frac{c}{2} \right]$.

We first find the candidates of maximizer, i.e. the critical points x^* .

$$f'(x) = \frac{c}{2} - 2x.$$

So $f'(x^*) = 0$ if and only if $x^* = \frac{c}{4}$.

The critical points are $x^* = 0$, $x^* = \frac{c}{4}$ and $x^* = \frac{c}{2}$.

Since f is continuous on a closed bounded interval domain $\left[0, \frac{c}{2} \right]$, by the EVT, the maximum (and minimum) exists and occurs at one of the critical points.

Since $f(0) = f\left(\frac{c}{2}\right) = 0$ and $f\left(\frac{c}{4}\right) = \frac{c^2}{16} > 0$, the maximum is obtained at $x = \frac{c}{4}$.

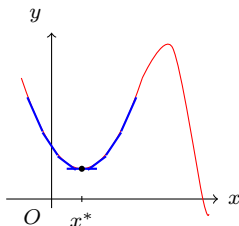
So, $x = \frac{c}{4}$ and $h = \frac{c}{4}$ is the optimal solution and the maximum area is $\frac{c^2}{16}$.

SECOND DERIVATIVE TEST

If f is twice differentiable, we can check whether a critical point gives a **local** maximum or a **local** minimum by using the **second derivative test**. However, the second derivative test **cannot** conclude whether a critical point gives a global maximum or a global minimum, and further analysis is required.

PROPOSITION (SUFFICIENT CONDITION FOR LOCAL MINIMA)

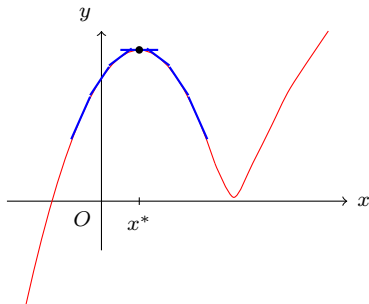
Suppose f is differentiable and $f''(x^*)$ exists, then $f'(x^*) = 0$ and $f''(x^*) > 0 \Rightarrow (x^*, f(x^*))$ is a local minimum.



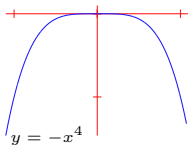
SECOND DERIVATIVE TEST

PROPOSITION (SUFFICIENT CONDITION FOR LOCAL MAXIMA)

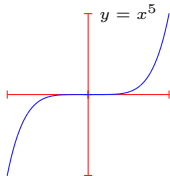
Suppose f is differentiable and $f''(x^*)$ exists, then $f'(x^*) = 0$ and $f''(x^*) < 0 \Rightarrow (x^*, f(x^*))$ is a local maximum.



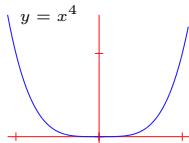
THE SECOND DERIVATIVE TEST - EXAMPLE



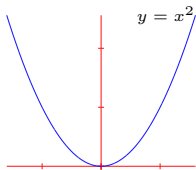
$f'(0) = 0$ and $f''(0) = 0$, so we can't apply the 2nd derivative test, though $(0,0)$ is a local maximum.



$f'(0) = 0$ and $f''(0) = 0$, so we can't apply the 2nd derivative test. $(0,0)$ is neither a local maximum nor minimum.



$f'(0) = 0$ and $f''(0) = 0$, so we can't apply the 2nd derivative test, though $(0,0)$ is a local minimum.



$f'(0) = 0$ and $f''(0) = 2 > 0$. $(0,0)$ is a local minimum by the 2nd derivative test.

ACTIVITY 3

Find the local and global extrema of the function

$$\begin{aligned} f : [-1, 4] &\longrightarrow \mathbb{R} \\ x &\longmapsto 3x^4 - 16x^3 + 18x^2, \end{aligned}$$

if they exist. Sketch its graph without using any graphing tools.

ACTIVITY 3: SOLUTION

$$\begin{aligned} f'(x) &= 12x^3 - 48x^2 + 36x \\ &= 12x(x^2 - 4x + 3) = 12x(x - 1)(x - 3). \end{aligned}$$

So $f'(x) = 0 \Leftrightarrow x = 0, 1$ or 3 .

Also, $f''(x) = 36x^2 - 96x + 36$.

$$f''(0) = 36 > 0 \quad \Rightarrow \quad \text{local minimum at } x = 0.$$

$$f''(1) = -24 < 0 \quad \Rightarrow \quad \text{local maximum at } x = 1.$$

$$f''(3) = 72 > 0 \quad \Rightarrow \quad \text{local minimum at } x = 3.$$

At the left endpoint $x = -1$, $f'(-1) = -96 < 0$, so the function is decreasing at $x = -1$ and there's a local maximum at $x = -1$. At the right endpoint $x = 4$, $f'(4) = 144 > 0$, so the function is increasing at $x = 4$ and there's a local maximum at $x = 4$.

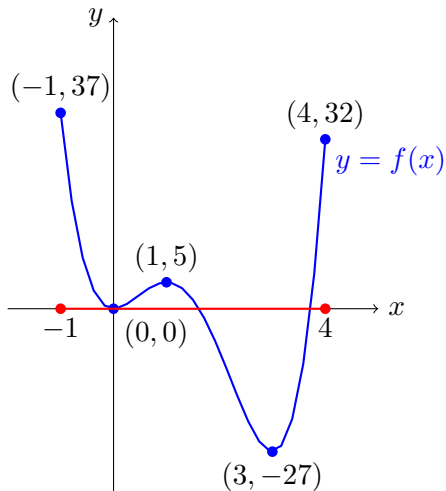
ACTIVITY 3: SOLUTION

Since f is continuous over the bounded closed interval $[-1, 4]$, by the EVT, f has global extrema and they occur at the critical points.

By comparing the values $f(-1) = 37$, $f(0) = 0$, $f(1) = 5$, $f(3) = -27$ and $f(4) = 32$, we conclude that f has a global maximum at $x = -1$ and a global minimum at $x = 3$.

To sketch the graph of $y = f(x)$, we use the first derivative to determine the intervals of increasing and decreasing. Since $f'(x) < 0$ for $x \in [-1, 0) \cup (1, 3)$, f is decreasing on $[-1, 0) \cup (1, 3)$. Also, $f'(x) > 0$ for $x \in (0, 1) \cup (3, 4]$, f is increasing on $(0, 1) \cup (3, 4]$. If we want to sketch the precise convexity of the graph, we will need to use the second derivative.

ACTIVITY 3: SOLUTION



CONVEX/CONCAVE FUNCTIONS

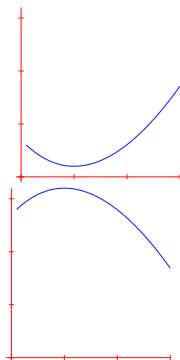
Lastly, you can also apply the following theorems to conclude that a local extremum is a global extremum.

THEOREM

If f is a convex function **on an interval domain**, then a local minimum of f is in fact its global minimum.

THEOREM

If f is a concave function **on an interval domain**, then a local maximum of f is in fact its global maximum.



⚠ Do not confuse the theorems above with the second derivative test. To prove a function is convex (resp. concave) over an interval I , you need to show $f''(x) \geq 0$ (resp. ≤ 0) for **all** $x \in I$. However, for the second derivative test, you only need to show $f''(x_0) > 0$ or < 0 , where $x = x_0$ is a critical point such that $f'(x_0) = 0$.

CONVEX/CONCAVE FUNCTIONS

E.g.: For the function $f(x) = x^4 + 2x^3 + 6x^2$ with domain $\mathbb{R}$,

$$f'(x) = 4x^3 + 6x^2 + 12x$$

and

$$f''(x) = 12x^2 + 12x + 12 = 12(x^2 + x + 1) > 0$$

(because the discriminant < 0), thus f is convex. At critical points,

$$f'(x^*) = 4(x^*)^3 + 6(x^*)^2 + 12x^* = 0$$

$$\Leftrightarrow 2x^*(2(x^*)^2 + 3x^* + 6) = 0$$

$$\Leftrightarrow x^* = 0,$$

because the discriminant of $2(x^*)^2 + 3x^* + 6$ is < 0 . Since $f''(x) > 0$ for all $x \in \mathbb{R}$, so $f''(0) > 0$ and thus $x = 0$ is a local minimizer. Furthermore, f is convex over $\mathbb{R}$, hence the local minimum at $x = 0$ is also a global minimum.

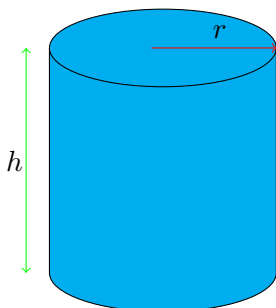
GENERAL STRATEGY

1. Convert the word problem to a math problem. Identify the objective function f and the decision variable(s).
2. Use the given relationships to manipulate f to become a function of a single variable x (only for this course).
3. Find the domain D of f , i.e. the set of all feasible x 's.
4. Find all the critical points x^* , they are the candidates to maximizers and minimizers.
 - (a) If f is continuous and D is a bounded closed interval, the EVT guarantees that the global extrema exist. Compute and compare the values of f at all the critical points. The largest is the global max, the smallest is the global min.
 - (b) If f is convex and D is an interval, every local min is global.
 - (c) If f is concave and D is an interval, every local max is global.
 - (d) You can always sketch the graph of f with the help of f' and identify the global extrema from it.

ACTIVITY 4

You are a manufacturer of cylindrical soup cans, and one day you receive a very large order for cans of a specified volume $V_0 > 0$. What dimensions (in terms of V_0) will minimize the total surface area of such a can, and therefore the amount of metal needed to manufacture it?

ACTIVITY 4: SOLUTION



We want to minimize

$$A = 2\pi r^2 + 2\pi r h.$$

By using the volume relationship

$V_0 = \pi r^2 h \Rightarrow h = \frac{V_0}{\pi r^2}$, the optimization problem is to minimize

$$A(r) = 2\pi r^2 + \frac{2V_0}{r}, \text{ for } r > 0.$$

Since $A'(r) = 4\pi r - \frac{2V_0}{r^2}$,

solving $A'(r^*) = 0$ gives $r^* = \sqrt[3]{\frac{V_0}{2\pi}}$. The interval of r is $(0, \infty)$, which is not a closed interval, so we cannot apply the Extreme Value Theorem here. As $A''(r) = 4\pi + \frac{4V_0}{r^3} > 0$ for all $r > 0$, $A(r)$ is a convex function over the interval $(0, \infty)$, hence the local minimum at $r^* = \sqrt[3]{\frac{V_0}{2\pi}}$ must be the global minimum.

ACTIVITY 4: SOLUTION

Alternatively, we see that $A'(r) < 0$ for $r < r^*$ and $A'(r) > 0$ for $r > r^*$, so the function $A(r)$ decreases from $r = 0$ to $r = r^*$ and then increases from $r = r^*$ to $r \rightarrow \infty$, thus r^* must be a global minimizer.

Since $h = \frac{V_0}{\pi r^2}$, so the optimal height is

$$h^* = \frac{V_0}{\pi (r^*)^2} = r^* \frac{V_0}{\pi (r^*)^3} = r^* \frac{V_0}{\pi \frac{V_0}{2\pi}} = 2r^* = 2\sqrt[3]{\frac{V_0}{2\pi}} = \sqrt[3]{\frac{4V_0}{\pi}}.$$

ACTIVITY 5

At a selling price of \$1.50, a door-to-door salesperson can sell 500 cans of soup that costs 70¢ each to manufacture. For every cent that the salesperson lowers the price, the number of cans sold can be increased by 25 units. What selling price will maximize the total profit?

ACTIVITY 5: SOLUTION

Let $x :=$ decrease in price of a can (in ¢) and so $x \in [0, 80]$.
We want to maximize $P(x) :=$ profit of the salesperson

$$P(x) = (80 - x)(500 + 25x),$$

$$P'(x) = 1500 - 50x.$$

$P'(x) = 0$ at $x = 30$ and $P(30) = 62500$. Also, at the end points, we have $P(0) = 40000$ and $P(80) = 0$. So by the Extreme Value Theorem, the maximum profit is $P(30)$ and the

best price is $\$1.50 - \$0.30 = \$1.20$

Alternatively, since $P''(x) = -50 < 0$, so $P(x)$ is concave over the interval $[0, 80]$. The local maximum at $x = 30$ (because $P'(30) = 0$ and $P''(30) < 0$) is a global maximum.

OTHER APPLICATIONS OF DIFFERENTIATION

Differentiation has many other useful applications. You will learn the following applications in Calculus in term 1.

- L'Hôpital's rule - computing limits of indeterminate forms using differentiation.
- Newton's method - a numerical method of finding roots of differentiable functions.
- Taylor polynomials and Taylor series - approximating analytic functions by polynomials, expressing analytic functions as power series (polynomials of infinite degree).

OUTLINE

- ① Graphs and Derivatives
- ② Optimization and Derivatives
- ③ Summary

SUMMARY

- Graphs and Derivatives
 - increasing and decreasing.
 - graphs and derivatives.
- Optimization and Derivatives
 - formulation of optimization problems: objective function, decision variable, feasible region.
 - Fermat's theorem.
 - the extreme value theorem.
 - critical points.
 - second derivative test.
 - global extrema of convex/Concave functions.

RESOURCES

References:

- George F. Simmons. Calculus with Analytic Geometry (2nd edition) sections 4.1-4.4.
- Hughes-Hallett. Calculus (6th edition) sections 4.1-4.3.